

Reflect Points Across Axes

Lesson 9-7

Name: _____

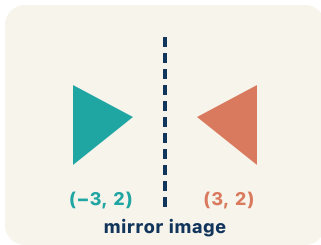
Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

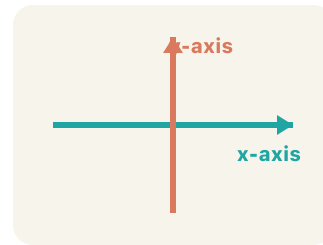
Picture first, then the word, then a plain-language meaning. Say each word out loud.



$(3, 2)$ reflected over the y -axis becomes $(-3, 2)$ — x changes sign, y stays

Reflection

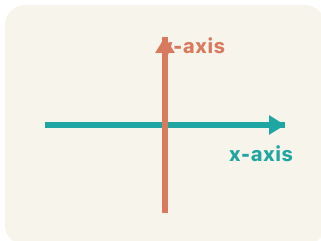
Write the definition:



$(4, 3) \rightarrow (4, -3)$: the y flips from $+3$ to -3 , like folding the paper along the horizontal line

x-axis

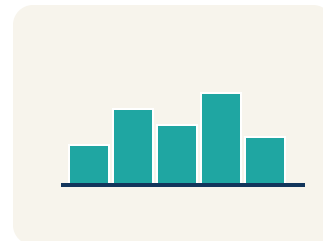
Write the definition:



$(4, 3) \rightarrow (-4, 3)$: the x flips from $+4$ to -4 , like folding the paper along the vertical line

y-axis

Write the definition:

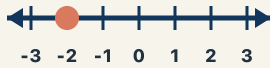


A butterfly's wings are symmetric — fold it down the middle and both sides match

Symmetry

Write the definition:

...-2, -1, 0, 1, 2...



..., -3, -2, -1, 0, 1, 2, 3, ...

Integer

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can reflect points across the x-axis and y-axis on the coordinate plane.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Reflection

x-axis

y-axis

Symmetry

Integer

1 A flip of a point or shape over an axis to make a mirror image is a _____.

2 The horizontal number line on the coordinate plane is the _____.

3 The vertical number line on the coordinate plane is the _____.

4 When two halves are mirror images of each other, the figure has _____.

5 A whole number or its opposite, with no fraction part, is an _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: What is the reflection of $(4, -3)$ over the x-axis?

- A. $(-4, -3)$
- B. $(4, 3)$
- C. $(-4, 3)$
- D. $(3, -4)$

Step 1 Reflecting over the x-axis changes the sign of the y-coordinate.

Step 2 The x stays the same (4), and the y changes from -3 to 3.

Step 3 The answer is $(4, 3)$.

 **Answer:** B. $(4, 3)$

 We do — together

Solve this one with your class using the same steps.

Problem: What is the reflection of $(-2, 6)$ over the y-axis?

- A. $(2, 6)$
- B. $(-2, -6)$
- C. $(2, -6)$
- D. $(6, -2)$

Step 1 _____

Step 2 _____

Step 3 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: Which point is on the y-axis?

- A. $(0, 5)$
- B. $(5, 0)$
- C. $(3, 3)$
- D. $(-2, 4)$

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. A buried chest is marked at $(-5, 3)$. Vega folds the map along the y -axis to find its mirror twin. Where is the twin?

- A. $(5, 3)$
- B. $(-5, -3)$
- C. $(5, -3)$
- D. $(3, -5)$

Show your work:

2. Vega reflects a palm-tree marker at $(8, -6)$ across the x -axis. Which coordinate changes its sign?

- A. The y -coordinate
- B. The x -coordinate
- C. Both coordinates
- D. Neither coordinate

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A triangle has vertices at $A(1, 2)$, $B(4, 2)$, and $C(4, 6)$. If you reflect the entire triangle over the y -axis, what are the new vertices? What do you notice about the size and shape of the reflected triangle compared to the original?

Sentence starter: The reflected vertices are $A'(\underline{\hspace{1cm}})$, $B'(\underline{\hspace{1cm}})$, $C'(\underline{\hspace{1cm}})$. I found each by $\underline{\hspace{1cm}}$. The reflected triangle is $\underline{\hspace{1cm}}$ compared to the original because reflections $\underline{\hspace{1cm}}$.

Show your work:

Reflect — Exit Ticket

What is the reflection of $(-5, 2)$ over the y -axis?

- A. $(-5, -2)$
- B. $(5, -2)$
- C. $(5, 2)$
- D. $(2, -5)$

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Reflection
2. **Notes 2:** x-axis
3. **Notes 3:** y-axis
4. **Notes 4:** Symmetry
5. **Notes 5:** Integer
6. **We Try (notes):** A. (2, 6) — *Reflecting over the y-axis changes the sign of the x-coordinate. x goes from -2 to 2, and y stays 6. The answer is (2, 6).*
7. **You Try (notes):** A. (0, 5) — *Points on the y-axis have x-coordinate = 0. (0, 5) is on the y-axis.*
8. **Try It 1:** A. (5, 3) — *Reflecting across the y-axis flips the sign of the x-coordinate only. (-5, 3) becomes (5, 3).*
9. **Try It 2:** A. The y-coordinate — *Across the x-axis the rule is $(x, y) \rightarrow (x, -y)$, so only the y-coordinate flips sign. (8, -6) becomes (8, 6).*
10. **Exit Ticket:** C. (5, 2) — *Reflecting over the y-axis changes the sign of the x-coordinate. x goes from -5 to 5, and y stays 2. The reflection is (5, 2).*